Do Now · retrieval

Lesson 047 · Two- and Three-dimensional Arrays in C#

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

How does index 0 work in C# arrays?

Q2

What is a property on a C# object?

Q3

What is the distinction between an array reference and an object instance?

Q4

A zero-indexed array a has 4 rows and 3 columns. Every element starts at 0. A loop runs r=0 to 3 inclusive and sets a[r,2]=r+1. Determine the final values in the last column, from top to bottom.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — A rectangular C# 2D array uses [row,column]. Example seats has 2 rows and 3 columns.

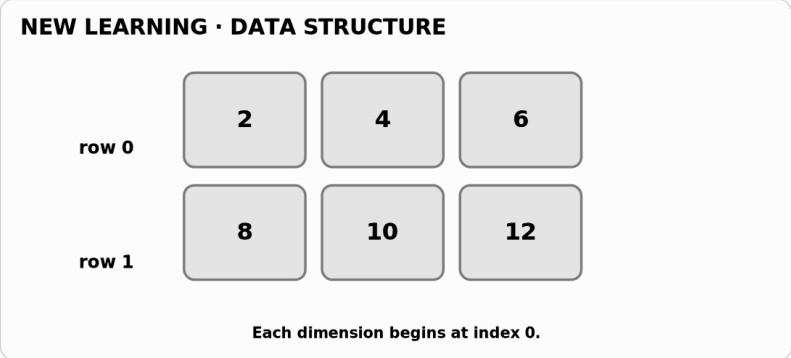
Core idea 2

Explain — A 3D array uses [layer,row,column]. new int[2,3,4] holds 24 integers; its last valid index is [1,2,3].

Core idea 3

Apply — Each dimension starts at zero; distinguish [1,2,3] from a single flat offset and never exceed a dimension's length.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

dimension — An indexing axis of an array.

row — First index in 2D example.

column — Second index in 2D example.

voxel — 3D grid element.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

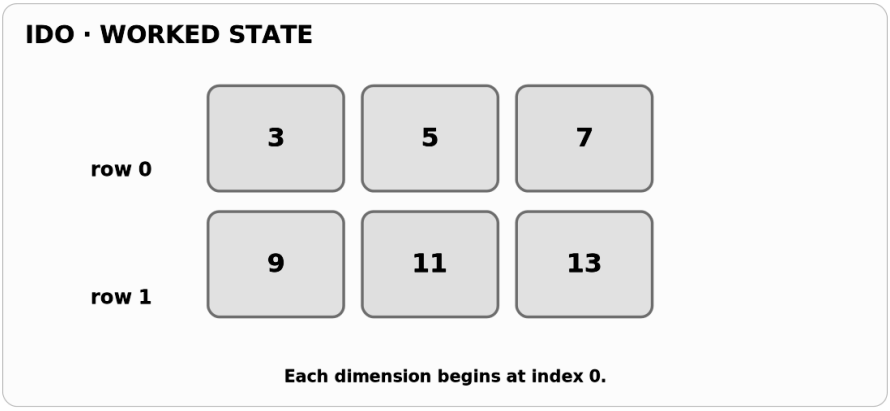
Problem

Find seats[0,2] in {{3,5,7},{9,11,13}}.

Follow the worked method

1. Trace and justify — First row, third column gives 7. The column index 2 is valid because there are three columns.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Trace and justify

Output 14: second layer, first row, third column. Six values per layer, twelve total.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Find a[1,1] in {{2,4,6},{8,10,12}} and show indexes. [4]

Question 2

A 3D array has sizes [2,3,4]. Calculate items and last index triple. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Write two nested C# loops that print all values in int[,] a of 3 rows and 2 columns, explaining order. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In int[,,] x=new int[2,3,4], x[2,0,0] is valid.

Use key vocabulary from memory

1. Explain dimension in your own words.

2. Explain row in your own words.

3. Explain column in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.